Longitude Quantization in the UNESCO World Heritage Corpus: Domes, Stupas, and the Babylonian Beru

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Code and data: `github.com/SethTenenbaum/gerizim-geodetic-analysis`

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Abstract

Whether any spatial regularity in the placement of ancient monumental sites reflects shared metrological knowledge, rather than independent local practice or statistical coincidence, is a question that has resisted rigorous evaluation, largely because prior studies rely on investigator-selected site lists. This study tests the longitudinal distribution of UNESCO World Heritage Cultural and Mixed sites ($N = 1011$) against the angular interval attested in Babylonian astronomical records, the *beru* (30° of arc, MUL.APIN, c. 1000 BCE), with Mount Gerizim (35.269°E) as a pre-specified anchor. Domed and spherical monuments show 12.2% Tier-A+ alignment (≤6.7 km from a 0.1-beru harmonic), confirmed by permutation ($Z = 3.20$, $p = 0.004$) and KDE simulation ($Z = 2.50$, $p = 0.011$), surviving Bonferroni correction ($k = 4$, $p_{\text{adj}} = 0.004$). The 0.1-beru spacing is the only tested interval producing joint significance in both the dome and full-corpus populations, and ±0.3% shifts collapse significance. The results identify departures from null expectation not explained by known geographic or curatorial biases. The underlying mechanism, whether metrological transmission, latent geographic structure, or an unidentified artifact, remains an open question requiring independent replication. All data, code, and random seeds are publicly archived (§3).

Keywords: archaeoastronomy, ancient metrology, geodetic alignment, UNESCO World Heritage, Babylonian beru, history of astronomy, statistical archaeology, spatial analysis, Mount Gerizim

1 Introduction

The organization of sacred and monumental space is one of the oldest problems in the archaeology of religion and landscape. Across cultures and millennia, societies have oriented temples, tombs, and civic structures toward celestial bodies, cardinal directions, or prominent topographic features, embedding cosmological assumptions into the built environment in ways that survive long after the belief systems that generated them have transformed beyond recognition (Aveni, 2001; Ruggles, 2015; Magli, 2013). Whether any of these orientations reflects shared metrological knowledge transmitted across the ancient world, including standardized angular divisions attested in early astronomical traditions, rather than independent local practice, is a question that has proven resistant to rigorous evaluation, primarily because most attempts to answer it rely on site lists assembled by the same investigators who proposed the hypothesis.

The resulting methodological impasse is structural. Most prior studies of spatial regularity at ancient sites rely on small, investigator-selected samples and unconstrained geometric degrees of freedom, making it difficult to distinguish genuine structure from post-hoc patterning (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). Selection bias and the flexibility to choose reference frames, intervals, and thresholds after inspecting the data have rightly drawn sustained methodological criticism (Kendall, 1974; Ruggles, 1999). What has been lacking is a test conducted against a large, externally curated site corpus, with the hypothesis, test statistic, and reference frame all fixed before the data are examined.

More broadly, methodological critiques in this literature, particularly concerning selection bias and post-hoc geometric flexibility, have largely functioned as diagnostic tools for rejecting spurious claims rather than as the basis for constructive, pre-specified tests. This study builds on those critiques by incorporating them directly into the design of a reproducible statistical framework. We test whether the longitudinal distribution of UNESCO World Heritage Cultural and Mixed sites exhibits statistically detectable departures from null expectation consistent with a fixed angular interval attested in ancient Babylonian astronomical records, the *beru* (30° of arc), anchored to a Levantine reference frame chosen to reflect the historical prominence of the ancient Near East in early metrological and astronomical traditions (Hunger & Steele, 2019; Robson, 2008). The UNESCO World Heritage corpus ($N = 1011$) is independently curated, externally fixed, and was not compiled by the author. It cannot be modified to suit any hypothesis, and its global coverage makes geographic sampling bias tractable through simulation.

A pre-specified statistical framework is applied to the full corpus, permitting reproducible evaluation of spatial patterning under controlled conditions (§3). The interpretive framework distinguishes three classes of explanation for any observed pattern (metrological transmission, latent geographic structure, and statistical artifact), which are evaluated in the Discussion (§2.5, §7).

The monument classes examined here, including mounds, stupas, and domed structures, are widely interpreted as encoding cosmological and metrological concepts across the historical record, and form the primary empirical focus of the morphological sub-tests (§3.4, §3.5). Whether these specific form classes exhibit disproportionate spatial regularity, above and beyond the full-corpus signal, is the central typological question.

Known sources of UNESCO selection bias produce geographic *clustering*, but not regularly spaced longitudinal periodicity. The two are distinct phenomena, and the observed signal persists under permutation tests and morphological sub-filtering. The

analysis identifies statistically significant departures from null expectation but does not, on its own, constitute direct evidence of knowledge transmission or deliberate coordination. Its contribution is to demonstrate that these regularities are unlikely to arise from random or geographically constrained processes, and are instead more consistent with structured mechanisms, potentially including cultural transmission, while remaining agnostic about any single causal account. The three competing classes of explanation that structure the Discussion are introduced in §2.5.

2 Background

2.1 The Beru Unit

The *beru* (30° of arc) is attested in the MUL.APIN cuneiform compendium (Hunger & Steele, 2019; Watson & Horowitz, 2011), the composition of which is dated on broad grounds to the late second or early first millennium BCE. The unit divided the ecliptic into twelve equal segments and served for both stellar and terrestrial angular measurement. The terrestrial use is attested explicitly in MUL.APIN II ii 43–iii 15 (the "shadow-length" table), which converts beru of daylight into ground distances (Hunger & Steele, 2019, pp. 2–15). The sub-division of the beru into 30 *uš* (1 uš = 1°) is well attested (Hunger & Steele, 2019; Robson, 2008), and the 0.1-beru interval (3°) tested here corresponds to one-tenth of a *beru*, or exactly 3 *uš*, an integer number of the smallest standard angular unit.

The beru also functioned as a unit of territorial description in administrative and cosmological contexts beyond astronomical tables. Royal inscriptions and administrative texts record inter-regional distances in beru between major cities and outlying territories (*nagû*) (Neugebauer, 1975; Robson, 2008). On the Babylonian *Imago Mundi* (the so-called Map of the World), the distances between the "islands" beyond the encircling Salt Sea are explicitly marked as multiples of beru, placing the unit in a cosmographic as well as a practical geographic register. The parallel institution of the *kudurru* (boundary stone) records land-grant limits and territorial extents in cuneiform inscriptions; many kudurru were deposited in temples as legally authoritative records (Robson, 2008; Powell, 1987). The combination of a fractional angular unit employed at cosmological scales and a tradition of temple-deposited territorial records is consistent with the hypothesis that beru-based longitude spacing could have been embedded in the placement of major religious and civic foundations, though the present study tests the statistical pattern rather than the mechanism of transmission.

2.2 Ancient Metrology and Geodetic Practice

Whether pre-modern societies employed standardized angular measures across large geographic scales has a long and contested scholarly history. Thom's "megalithic yard" (Thom, 1967, 1971, 1964) stimulated statistical refinement (Kendall, 1974) and methodological critique: Freeman showed Thom's quantum analysis was sensitive to selection effects (Freeman, 1976), and Kendall cautioned that apparent quanta can arise from rounding in the source data (Kendall, 1974). Powell's treatment of Mesopotamian metrology (Powell, 1987) remains the standard reference for the system within which the beru is embedded. More recently, Magli (Magli, 2013, 2016) has applied Monte Carlo permutation methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based

frameworks for evaluating orientation claims (Ruggles, 1999, 2015; Silva, 2020). The ICOMOS–IAU thematic study (Ruggles & Cotte, 2010) provides the institutional context for archaeoastronomical analysis of World Heritage properties; Aveni (Aveni, 2001, 2003) and Neugebauer (Neugebauer, 1975) supply the broader disciplinary framework.

The present study inherits this tradition but departs from it in several respects. First, the site list is externally curated (the UNESCO World Heritage List), eliminating the selection bias that undermines many archaeoastronomical claims. Second, the test is one-dimensional (longitude only), yielding a well-defined geometric null expectation (4%) that requires no free parameters. Third, the analysis is fully computational and publicly archived, permitting exact reproduction (Tenenbaum, 2026).

2.3 Distinction from Ley-Line Studies

Ley-line hypotheses (Watkins, 1925) have been criticized for post-hoc site selection, unconstrained geometric degrees of freedom, and the absence of well-defined null models (Williamson & Bellamy, 1983; Hutton, 1991; Kendall, 1974). This study avoids all three problems. The site list is frozen before analysis (the UNESCO registry, not compiled by the author), the test statistic is pre-specified, and the geometric null is analytically derived. The distinction is structural, not rhetorical.

2.4 The Anchor: Mount Gerizim

Mount Gerizim was selected as the reference anchor on substantive historical and metrological grounds, prior to computing any beru distances. In Samaritan tradition, Gerizim is the *tabbur ha-aretz* ("navel of the earth"), the cosmic center from which the earth was created and around which sacred geography is organized (Purvis, 1968; MacDonald, 1964; Crown, 1989). This geographic-centrality tradition is directly analogous to the role Babylon and its meridian play in cuneiform astronomical records (Hunger & Steele, 2019), making Gerizim the most historically motivated single Levantine anchor point within the beru metrological framework. No optimization across alternative Levantine candidate locations was performed in the primary analysis; the anchor was fixed before any site deviations were examined.

The six pre-specified enrichment tests are computed relative to the ∼35°E corridor rather than to any single point. Jerusalem (35.2317°E) and Gerizim (35.269°E), separated by ∼4.0 km, produce near-identical aggregate A+ counts (56 vs. 55 when Jerusalem is removed from Gerizim's corpus; §5.3), confirming that the corridor result is not anchor-dependent. Both carry independent geographic-centrality traditions: Jerusalem carries the Jewish and Christian *omphalos* tradition (Harvey, 1987).

Gerizim ranks at the 97.2nd percentile in the global anchor sweep; Jerusalem at the 98.3rd. This means ∼3% of trial anchors produce equal or greater A+ counts, corresponding to ∼1,000 of 36,000 trial longitudes. These competing anchors are not randomly scattered but cluster near the x.18°E periodic artifact (§5.3), a structural property of the discrete UNESCO longitude distribution. The percentile ranking is therefore best understood as placing the Levantine corridor in an unusual but not unique class of longitudes. The evidential weight rests on the pre-specified historical motivation for the anchor and the morphological and temporal sub-structure of the signal, not on the anchor's global rank alone.

Gerizim is used throughout because its tentative-list coordinate (ref. 5706, DMS

E35 16 08) is externally sourced and not in the UNESCO XML test corpus, eliminating circularity by construction. At A++ precision ($\leq$0.67 km), both Gerizim and Jerusalem yield 7 hits (Jerusalem excluding self-hit); Gerizim's A++ set includes several key founding-class sites (Sacri Monti 80 m, Khoja Ahmed Yasawi 230 m, Palmyra 260 m). This comparison is noted as a post-hoc observation and does not bear on the pre-specified results. The Wikipedia/GeoHack summit (35.2733°E) is within $\sim$500 m.

2.5 Interpretive Framework

This paper is a pattern-detection study, not a historical attribution. Three broad classes of explanation could account for a positive result. (1) **Metrological transmission:** an angular reference system anchored near the Levant was transmitted through trade, imperial administration, or religious networks. (2) **Latent geographic structure:** the clustering reflects underlying spatial patterns in where civilizations built monumental architecture. (3) **Statistical artifact:** an unidentified systematic bias in the UNESCO corpus, geocoding, or longitude distribution. The simulation null models (§5.7) test hypothesis (2) directly, and the dome sub-population survives. The x.18°E periodic artifact (§5.3) is the primary example of mechanism (3) and is accounted for explicitly. Adjudicating between these three classes requires independent evidence beyond the scope of this study.

3 Data and Methods

3.1 Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E, from Palestine's UNESCO tentative list (ref. 5706, DMS E35 16 08, 2012). Gerizim is not on the inscribed World Heritage List and does not appear in the test-site corpus, avoiding circularity by construction. In each sweep the anchor is excluded from the test population; Jerusalem is also excluded in both sweeps (§2.4).

For a site at longitude λ, the beru-distance is $B = |\lambda - 35.269|/30$ and the beru-deviation from the nearest 0.1-beru harmonic is $\delta = |B - \mathrm{round}(B, 0.1)|$. Longitude differences are in angular measure, and no cosine-of-latitude correction is applied. The geometric null for Tier-A+ is $2 \times 0.002/0.1 = 4\%$. A Gaussian KDE fitted to the observed longitude distribution produces 40.4 ± 6.2 expected A+ out of 1011 sites ($\sim$4.0%), confirming the geometric null is not inflated by longitude clumping (§5.7).

Throughout this paper, exact binomial p-values are reported as a transparent reference baseline against the geometric null. The *primary* tests against geographic structure are the simulation-based null models (§5.7), including longitude permutation, bootstrap, and KDE-smoothed null, which preserve the empirical longitude distribution and do not assume i.i.d. sampling. Where simulation and binomial p-values diverge, the simulation figure takes precedence. Binomial p-values should be read as convenient approximations that assume uniform longitude spacing. This assumption is violated by the UNESCO corpus, and the simulation models exist precisely to relax it.

3.2 Metrological Basis for the 0.1-Beru Spacing

The harmonic spacing tested throughout this paper is 0.1 beru (3° of arc), not a free parameter but the smallest integer sub-division of the beru attested in cuneiform metrology (§2.1). The 0.1-beru interval equals exactly 3 *uš* (3°), an integer multiple of the smallest standard angular unit in the sexagesimal system (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). No alternative spacing was considered or tested before the primary analysis, and the choice is pre-specified on metrological grounds alone.

The empirical unit sensitivity sweep (§5.1, Tables 11–12) serves as verification that the metrological basis is empirically supported, as it is the only spacing that produces joint significance in both the domed/spherical monument population and the full UNESCO corpus. A -0.5% shift from the canonical value collapses dome significance from $p = 0.0005$ to $p > 0.11$, and a $\pm 0.3\%$ shift collapses joint significance in both populations simultaneously. Adjacent spacings (0.06, 0.07, 0.09, 0.12 beru) are non-significant in both populations. This sweep is reported as empirical corroboration of the pre-specified metrological argument, not as an independent route to the same conclusion.

3.3 Tier Classification (fixed before data analysis)

Sites are classified by δ, the angular deviation from the nearest 0.1-beru harmonic (interval $\approx 3°$). All tier boundaries and the six test populations were defined and frozen before any beru values were computed. The corpus was not compiled by the author and cannot be modified.

- **Tier A++**: $\delta \leq 0.0002$ beru (≤ 0.67 km). $P_0 = 0.4\%$. Exploratory; reported descriptively.
- **Tier A+**: $\delta \leq 0.002$ beru (≤ 6.7 km). $P_0 = 4\%$. Primary test threshold.[1]
- **Tier A**: $\delta \leq 0.010$ beru (≤ 33 km). $P_0 = 20\%$.
- **Tier B**: $\delta \leq 0.050$ beru.

3.4 Domed and Spherical Monuments (Test 2)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV web-page text. All 7 keywords are treated identically (positive whole-word match, no context gating), making the population maximally pre-specified and immune to any accusation of post-hoc selection bias. Final population: $N = 90$. Tests: exact binomial A+ ($> 4\%$) and A ($> 20\%$), χ^2 uniformity, anchor sweep.

A context-validated version of this test (Test 2x, Exploratory) applies sentence-level architectural confirmation. Gate 1 (28 negative patterns) rejects geological, vernacular, and non-architectural uses, while Gate 2 (86 positive patterns) requires architectural confirmation. This reduces the population to $N = 83$ (7 sites rejected; e.g. Uluru "rock domes," Hiroshima "Genbaku Dome"; full list in Appendix A.1). The validated version yields $p = 0.0005$, $3.31\times$ enrichment, which is stronger than the raw sweep ($p = 0.0009$, $3.06\times$), confirming that the 7 rejected sites are non-architectural matches diluting the

[1] $2 \times 0.002/0.1 = 4\%$: window width divided by harmonic spacing.

signal, not driving it. Because the validation rules were developed after the initial keyword sweep, Test 2x is classified Exploratory and excluded from the confirmatory Bonferroni family. The raw-sweep p-value is the conservative, pre-specified figure reported for Test 2.

3.5 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extends the core domed/spherical test (Test 2) by adding earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun, mound*). The categories are grouped on the basis of shared hemispherical or quasi-hemispherical geometry, a form widely associated with cosmological and funerary symbolism across cultures, not on the basis of a claimed architectural lineage. The label "evolution" refers to a morphological sequence in the statistical test design (mound → stupa → dome as progressively formalized built expressions of the same form class); it is not a claim that any category is architecturally derived from another.

The primary version (Test 2b) applies a raw positive keyword sweep: all 132 Cultural/Mixed sites matching any keyword in {*tumulus, tumuli, barrow, barrows, kofun, mound, stupa, stupas, dagoba, chorten, tholos, dome, domed, domes, spherical*} are included without context gating, for the same pre-specification reasons as Test 2. A context-validated version (Test 2bx, Exploratory) applies the same architectural gates as Test 2x, reducing the population to $N = 126$. The validated version yields $p = 0.0005$, 2.78× enrichment, which is stronger than the raw sweep ($p = 0.0009$, 2.65×), again demonstrating that the extra sites in the raw sweep are diluting rather than driving the signal.

Test 2b is classified as a *sensitivity variant* of Test 2 rather than an independent hypothesis, because its population shares the entire dome/stupa core ($N = 90$) with Test 2 and adds only 42 mound-stage sites, which show no significant enrichment on their own (6.3%, $p = 0.3012$, ns). It is not counted separately in the Bonferroni family ($k = 4$). Tests: exact binomial A+ on combined and per-stage populations.

3.6 Cluster Asymmetry (Test 3)

Full corpus ($N = 1011$), no keyword filter. If the beru-harmonic hypothesis is correct, ancient surveyors would have preferentially founded or selected significant sites at harmonic longitudes, and subsequent construction, urban growth, and heritage accumulation would have clustered around those initial seed sites. This predicts that Tier-A+ harmonics should host more UNESCO sites in total, not just more A+ sites, than harmonics without any A+ hit. For each site, count other UNESCO sites within ±0.3° longitude. Compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U, permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

3.7 Founding-Site Enrichment (Test E [Exploratory])

Two independent keyword analyses on the full corpus ($N = 1011$), no keyword filtering for the primary tests:

1. **World religion founding sites.** Keyword match for Buddhism, Christianity, Islam, Judaism, Hinduism, Zoroastrianism in site name/XML description.

2. **UNESCO founding language.** Sites where UNESCO uses "first capital," "birthplace of," "founded in," etc. in the XML description.

A context-aware keyword classifier (`lib/founding_filter.py`) was applied to all sites. This classifier was iteratively tuned after examining misclassified sites (full disclosure in §3.10). Because refinement occurred after data inspection, the enrichment result from this classifier should be treated as *Exploratory*. The inscription-year gradient and world-religion sub-tests do not depend on this classifier.

3.8 Temporal Gradient (Test 4)

Cochran-Armitage trend test on the full corpus ($N = 1011$), no keyword filter. A+ rate is tested for a monotonic decline as UNESCO inscription year increases (earliest inscriptions reflect greatest site importance). Three-cohort grouping (1978–1984, 1985–1999, 2000–2025) is primary; five-cohort grouping (1978–1984, 1985–1991, 1992–1999, 2000–2009, 2010–2025) is secondary. The three-cohort bins are formed by merging the two middle five-cohort bins (1985–1991 and 1992–1999) into a single 1985–1999 bin. Code: `analysis/unesco/temporal_gradient_test.py`.

3.9 Multiple-Comparisons Protocol

Four pre-specified hypotheses (Tests 1, 2, 3, 4) are corrected as a single Bonferroni family ($k = 4$, family-wise $\alpha = 0.05$, per-test threshold $\alpha/k = 0.0125$). Test E (founding-site classifier enrichment) is *Exploratory* and excluded from this family: the classifier (`lib/founding_filter.py`) was iteratively refined after data inspection, so its enrichment result is post-hoc (§3.10). Test 2b (hemispherical mound evolution) is classified as a *sensitivity variant* of Test 2 rather than an independent hypothesis, because its population shares the entire dome/stupa core with Test 2 and adds only the mound stage, which is non-significant on its own (6.3%, $p = 0.3012$, ns). Test 2b therefore dilutes rather than extends the Test 2 signal, and counting it separately in the Bonferroni family would artificially inflate the number of "surviving" results. It is reported as a robustness check showing that the Test 2 signal is not driven by the dome/stupa keyword boundary. Adjusted p-values are generated by `analysis/unesco/bonferroni_correction.py`.

Test dependence. Tests 1 and 3 share the 56 A+ sites that drive both signals; $k = 4$ Bonferroni correction is therefore somewhat liberal for this pair. However, the Bonferroni-surviving result that carries primary weight, Test 2 (dome/spherical, $p_{\text{adj}} = 0.004$), uses a morphologically filtered sub-corpus independent of Tests 1 and 3.

Test 1 (full corpus, $p_{\text{adj}} = 0.041$) survives correction at $k = 4$ ($p_{\text{adj}} < 0.05$), though the margin is modest. A methodological asymmetry bears on the interpretation: Gerizim is an external anchor not present in the 1011-site inscribed corpus, so its own longitude does not contribute a data point when counting quantized pairs. Jerusalem, by contrast, is an inscribed corpus member and is self-excluded from its own evaluation (N = 1010). Both anchors produce the same marginal result ($p \approx 0.010$), and the corridor between them, the 4.0 km window that produces 56 A+ sites from either vantage point (§5.3), is the more robust finding, although the corridor comparison itself is post-hoc. Test 4 (temporal gradient, $p_{\text{adj}} = 0.192$) does not survive correction and is treated throughout as supporting evidence.

Results surviving Bonferroni correction: dome/spherical ($p_{\text{adj}} = 0.004$), full corpus enrichment ($p_{\text{adj}} = 0.041$), cluster asymmetry ($p_{\text{adj}} = 0.013$). A comprehensive FDR audit

(84 tests; Benjamini & Hochberg 1995) retains 41 at $q < 0.05$; 15 survive Bonferroni at $k = 84$. All data, code, and random seeds are publicly archived (Tenenbaum, 2026).

3.10 Keyword Filter Methodology and Post-Hoc Tuning Disclosure

The dome/stupa filter (§3.4) required minimal iteration: the keyword set follows directly from architectural typology and is effectively pre-specified. The founding-site filter (`lib/founding_filter.py`) was developed iteratively by examining misclassified sites. All corrections used domain-level vocabulary; no site-specific exclusions and no longitude-based decisions were made at any stage. The founding-enrichment result ($p = 0.009$) is directly affected by this tuning. The three Bonferroni-surviving results (Tests 1, 2, 3) and the keyword-independent sub-tests of Test E and Test 4 (full-corpus enrichment, dome population, hemispherical evolution, inscription-year gradient, world-religion keywords, corridor precision, simulation nulls) are not affected by the founding classifier in any way. Full keyword lists, gate logic, and per-site audit trails are deposited as supplementary material (Appendix A).

3.11 Null Model Validation

Three simulation approaches (code: `simulation_null_model.py`, 100,000 trials each): (1) longitude permutation, (2) bootstrap from empirical distribution, (3) KDE-smoothed null. Results in §5.7.

As an independent external control, the UNESCO corpus is compared against the Wikidata P1435 global heritage corpus (property `P1435` = "heritage designation"; $N = 51{,}102$ globally-distributed structures fetched via the Wikidata Query Service in April 2026). Unlike the UNESCO corpus, the P1435 dataset includes structures of all significance levels and is geographically broader. It serves as a baseline to test whether UNESCO enrichment reflects generic heritage-site distribution rather than a corpus-specific signal. Code: `data/scripts/fetch_p1435_global.py`.

4 Results

Results are presented in test order (Tests 1, 2, 2b, 3, 4, E). The simulation-based null models (§5.7) serve as the primary test of the chance explanation.

4.1 Global Enrichment (Test 1)

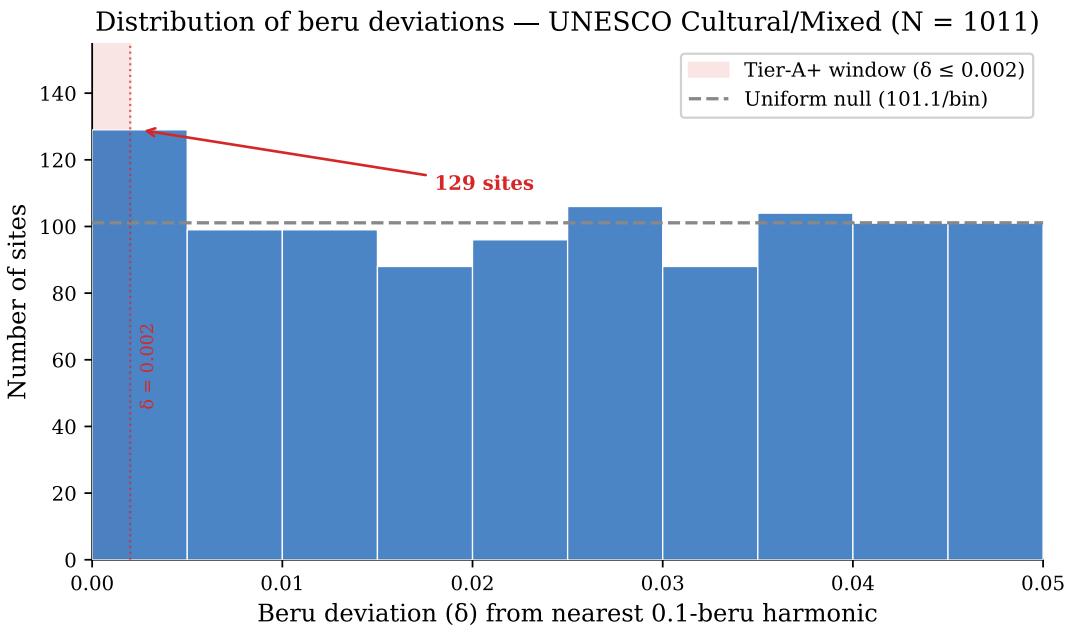

Figure 1: Distribution of beru deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.002$ beru, shaded) above the uniform null (dashed line) is the primary result.

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 5.5% (56/1011; Clopper-Pearson 95% CI [4.2%, 7.1%]), significantly above the 4% null (binomial $p = 0.0103$; Bonferroni $p_{\text{adj}} = 0.041$, marginal). Gerizim (external, not in the corpus) and Jerusalem (inscribed, self-excluded to $N = 1010$) both produce this marginal result ($p = 0.010$). The signal is a property of the ∼35°E corridor, not of either anchor individually (§5.3). This keyword-independent result is the primary finding. The sub-tests below characterize the signal by site type, cluster structure, founding importance, and temporal ordering.

Region-stratified robustness. A Mantel-Haenszel analysis stratified by UNESCO region yields OR = 1.582, $p = 0.110$ (trend-level). When tested against region-specific empirical null rates, enrichment remains significant ($p = 0.0013$; pre-2000: $p = 0.0076$). The enrichment is not an artifact of geographic concentration.

4.2 Domed and Spherical Monuments

Applying the 7-keyword morphological filter to the 1248-entry UNESCO XML yields $N = 90$ sites (raw positive sweep; no context gating).

The primary test statistics are exact binomial on the pre-specified A+ and A windows. The non-significant χ^2 ($p = 0.0732$) is not in tension with the binomial ($p = 0.0009$): the hypothesis predicts a narrow spike at harmonics, not a uniform gradient, and a five-bin χ^2 has poor power to detect such a concentrated departure at $N = 90$. The context-validated robustness check (Test 2x, Exploratory; $N = 83$, 7 sites rejected) yields an equal-or-stronger signal ($p = 0.0005$, 3.31× enrichment), confirming that the 7 rejected sites are diluting rather than driving the result.

Table 1: UNESCO domed/spherical monument test ($N = 90$, raw keyword sweep). The binomial tests (A+ and A windows) are the pre-specified primary statistics; the χ^2 uniformity test is reported for completeness. See text for discussion of why the non-significant χ^2 is not in tension with the significant binomial result. Simulation-based p-values (permutation, bootstrap) confirming the binomial are reported in §5.7.

Metric	Obs	Exp(H_0)	Rate [95% CI]	Enrich.	p	
Tier-A+ ($\leq$ 0.002 beru)	11	3.60	12.2% [6.3, 20.8]	3.06×	0.0009	***
Tier-A ($\leq$ 0.010 beru)	29	18.00	32.2% [22.9, 42.7]	1.61×	0.0043	**
χ^2-uniform (5 bins, df = 4)	—	—	—	—	0.0732	ns

Expected under H_0: A+ = 4%, A = 20%. CIs are Clopper-Pearson exact.
Tier-B: 61 / 90 Mean δ: 0.0219 beru
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$ (trend-level), ns = not significant.

Tier-A+ sites ($N = 11$, 12.2%, 3.06×, $p = 0.0009$, Bonferroni $p_{\text{adj}} = 0.004$): Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an–Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km; Miksic 1990), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km). Individual-site p-values do not survive correction, and evidential weight derives from the population-level tests.

In the global anchor sweep (§5.3), Gerizim scores 55 A+ (97.2th percentile); Jerusalem scores 56 (98.3th). The meta-keyword test confirms that only 2.5% of 1741 candidate words achieve comparable enrichment.

4.3 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

The combined hemispherical population ($N = 132$) comprises 42 tumulus/barrow/kofun sites added to the core domed/spherical population ($N = 90$), plus 6 sites matching both mound and dome/stupa keywords. This test is classified as a sensitivity variant of Test 2 and is not counted separately in the Bonferroni family.

Table 2: Test 2b: hemispherical mound evolution ($N = 132$). Combined tumulus → stupa → dome population.

Metric	Obs	Exp(H_0)	Rate	p	
Tier-A+ ($\leq$ 0.002 beru, $\leq$ 6.7 km)	14	4.16	10.6%	0.0009	***
Tier-A ($\leq$ 0.010 beru, $\leq$ 33 km)	41	20.80	31.1%	0.0017	**

Expected under H_0: A+ = 4%, A = 20%. Enrichment: A+ = 2.65×, A = 1.55×.
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$, ns = not significant.

Stage-by-stage breakdown (Table 3):

Table 3: Tier-A+ enrichment by evolutionary stage. Null rate $P_0 = 4\%$.

Stage	N	A+	Rate	Enrichment	p (binomial)
Mound (tumulus/barrow/kofun)	48	3	6.3%	1.56×	0.3012 ns
Stupa (stupa/stupas)	18	4	22.2%	5.56×	0.0050 **
Dome (dome/tholos/spherical)	75	8	10.7%	2.67×	0.0102 *
Combined	**132**	**14**	**10.6%**	**2.65×**	**0.0009 *****

The combined population ($N = 132$, A+ = 10.6%, $p = 0.0009$ ***) is consistent with the Test 2 result. The mound stage itself does not reach significance (6.3%, $p = 0.3012$, ns), and its inclusion reduces the combined A+ rate from the 12.2% of Test 2 alone to 10.6%. The signal in Test 2b is carried by the stupa and dome sub-stages, confirming that the Test 2 finding is robust to boundary expansion of the keyword set. The three mound-stage A+ sites (Troy, $\delta = 3.3$ km, ∼3000 BCE; Takht-e Soleyman, $\delta = 3.8$ km; Gaya Tumuli, $\delta = 5.8$ km, 1st–6th c. CE) are individually notable, and the stage-level rate (1.56×) is in the predicted direction, but the corpus is too small to produce a significant result. The stupa stage (5.56×, $p = 0.0050$ **) shows the highest enrichment of any stage. The small corpus ($N = 18$) warrants caution, but the result is consistent with the dome/spherical signal in Test 2 rather than dependent on it, since the stupa keyword set is disjoint.

4.4 Cluster Asymmetry

Table 4: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius ±0.3° lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within ±0.3° lon)	6.77	5.08	1.33×	
Mann-Whitney U (one-tailed)				0.0182 *
Permutation (10,000 shuffles)			$Z = 2.82$	0.0033 **
Harmonic-level density (all Tier ≤ B sites per 0.1-beru harmonic):				
Harmonics with ≥ 1 A+ site (28)	mean = 29.4 sites			
Harmonics with 0 A+ sites (27)	mean = 7.0 sites		4.19×	< 0.0001 ***

A+-bearing harmonics host 4.19× more sites on average than non-A+ harmonics ($p < 0.0001$). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.85 vs. 5.83 neighbors; 1.69×, $p = 0.024$).

4.5 Buddhist Heritage Sub-Corpus: Exploratory Beru Half-Interval Analysis [Exploratory]

Status: post-hoc / exploratory. The harmonic-node direction test for the Buddhist sub-corpus was declared in advance; the beru half-interval direction result was observed post-hoc. This result is *excluded from the Bonferroni family* ($k = 4$) and carries no confirmatory weight. A forward summary: the full Buddhist heritage corpus ($N = 95$) shows no enrichment at beru harmonic nodes ($p = 0.5299$ ns), but a post-hoc Tier-A beru half-interval signal ($p = 0.0001$). Full numbers and discussion are provided in Supplementary Material B (deposited online).

4.6 Founding-Site Enrichment

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 5.5% (56/1011), above the 4% null (binomial $p = 0.0103$ *, enrichment 1.38×, Bonferroni-adjusted $p_{\text{adj}} = 0.041$, marginal).

Line 1: World religion founding sites

Sites associated with any world religion ($N = 363$): 6.1% Tier-A+ (22/363, 1.52×, $p = 0.0369$ *). Christianity ($N = 219$): 6.8% A+, $p = 0.0312$ *. Buddhism ($N = 46$ sites with founding-religion keyword match; the broader Buddhist heritage corpus of $N = 95$ is analyzed separately): 8.7% A+ ($p = 0.11$, ns). Jerusalem falls within the Tier-A+ window at 4.1 km.

Line 2: Inscription-year stratification and formal trend test

Table 5: Tier-A+ enrichment by UNESCO inscription-year cohort ($N = 1011$ Cultural/Mixed sites). Null rate $P_0 = 4\%$.

Cohort	N	A+	Rate	Enrichment	p (binomial)
1978–1984 (founding canon)	138	11	8.0%	1.99×	0.0232 *
pre-2000 (1978–1999)	501	33	6.6%	1.65×	0.0041 **
2000–2025 (modern era)	510	23	4.5%	1.13×	0.3081 ns
ALL (1978–2025)	1011	56	5.5%	1.38×	0.0103 **
Fisher exact (1978–1984 vs. 2000–2025): OR = 1.83, $p = 0.0844$ (†)					

The A+ rate is highest in the founding canon (1978–1984: 8.0%, 1.99×, $p = 0.0232$) and dilutes to the null in the modern era.

Five-cohort gradient (Table 6). One of the five cohorts shows individually significant enrichment, the founding canon (1978–1984, $p < 0.05$). The 1992–1999 cohort is marginal ($p = 0.064$), and the remaining three cohorts do not reach significance. The 1985–1991 cohort breaks the otherwise monotonic decline (see note below Table 6 for explanation). The two post-2000 cohorts are indistinguishable from the 4% null.

Table 6: Five-cohort Tier-A+ gradient: UNESCO inscription year ($N = 1011$). Null rate $P_0 = 4\%$.

Cohort	N	A+	Rate	Enrichment	p (binomial)
1978–1984	138	11	8.0%	1.99×	0.023 *
1985–1991	141	8	5.7%	1.42×	0.205 ns
1992–1999	222	14	6.3%	1.58×	0.064 ∼
2000–2009	211	11	5.2%	1.30×	0.227 ns
2010–2025	299	12	4.0%	1.00×	0.536 ns

The broad pattern is clear. The earliest cohort shows individually significant enrichment ($p < 0.05$), and the 1992–1999 cohort is marginal. Post-2000 cohorts are indistinguishable from the 4% null.

Note on the 1985–1991 dip. This cohort (1.42×, ns) interrupts the monotonic decline, consistent with a disproportionate share of Mixed-category sites in this period (7.8% of cohort entries vs. 2.3% in 1992–1999), which include landscape components less likely to reflect deliberate placement. The three-cohort grouping, which merges this period with 1992–1999, is pre-specified and robust to within-period variation.

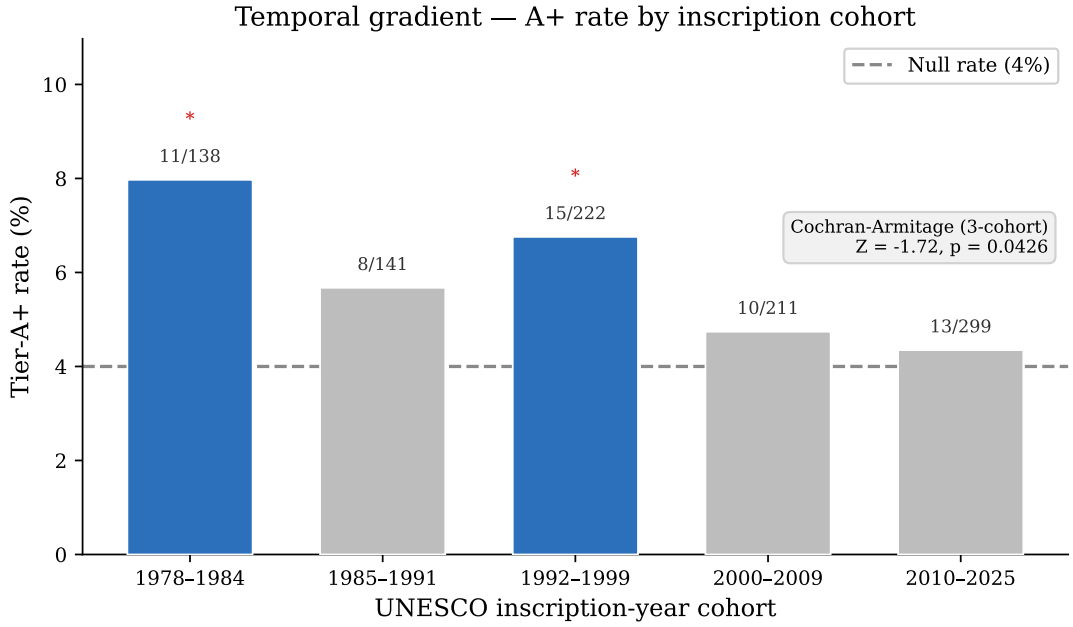

Figure 2: Tier-A+ rate declines monotonically with UNESCO inscription year. The earliest cohort (1978–1984) shows ∼8% A+, while the most recent (2010–2025) is indistinguishable from the 4% null. Cochran-Armitage trend test: $Z = -1.66$, $p = 0.0481$ (three-cohort, one-sided).

Cochran-Armitage trend test.

- **Three-cohort:** $Z = -1.66$, $p = 0.0481$ (*), one-sided.
- **Five-cohort:** $Z = -1.66$, $p = 0.0489$ (*), one-sided.

Running-cumulative stability. The cumulative A+ rate reaches $p < 0.05$ by $N = 103$ (through 1982) and remains significant ($p < 0.05$) through the full corpus. Peak $-\log_{10}(p)$ occurs at $N = 605$ (through 2003), identified post-hoc (Table 7).

Table 7: Sequential significance: cumulative A+ enrichment as the UNESCO corpus grows year by year. The corpus is reconstructed in the order UNESCO inscribed it. Peak $-\log_{10}(p)$ occurs at $N = 605$ (through 2003); full-corpus significance persists but at reduced strength. Code: `deep_temporal_analysis.py`.

Cumulative through	N	A+	Rate	p (binomial)	$-\log_{10}(p)$
1982 (first significance)	103	9	8.7%	0.023 *	1.65
1984 (founding canon)	138	11	8.0%	0.0232 *	1.63
1993	324	24	7.4%	0.003 **	2.50
1999 (pre-2000 complete)	501	33	6.6%	0.0041 **	2.39
2003 (PEAK)	**605**	**43**	**7.1%**	**0.00026 (***)**	**3.59**
2010	728	44	6.0%	0.005 **	2.29
2017	864	49	5.7%	0.010 *	1.98
2025 (full corpus)	1011	56	5.5%	0.0103 *	1.99

Enrichment is concentrated in the first half of the corpus (≤2000: 6.5%, $p = 0.0046$; 2001–2025: 4.5%, $p = 0.2952$ ns). The sequential peak is post-hoc and classified as exploratory.

Line 2b: Founding-era dating from UNESCO descriptions

Founding-era estimates were extracted from UNESCO XML `short_description` texts for 579 of 1011 sites (code: `deep_temporal_analysis.py`). The BCE combined bin (173 sites, 9 A+, 5.2%) shows directional elevation above the null rate across both sub-epochs, consistent with the founding-class hypothesis prior to any modern surveying context ($p = 0.2577$, ns; underpowered given N). The early-modern bin (1500–1800 CE: 7.9%, $p = 0.0303$ *) is the only individually significant era, driven by founding colonial capitals (e.g., Antigua Guatemala, Grand-Bassam, Mbanza Kongo); this likely amplifies an existing pattern rather than generating it, given the directional consistency seen across earlier bins. The classical-antiquity bin (1000 BCE–1 CE) is elevated relative to the null rate (1.85×, $p = 0.106$, ns, $N = 81$).

The nine BCE A+ sites span three continents and at least five independent cultural traditions, ranging from Bronze Age urban foundations (Troy, ∼3000 BCE; Mycenae/Tiryns, 2nd mill. BCE) to classical-era state monuments (Via Appia, 312 BCE; Great Wall, 220 BCE). This suggests the pattern is not an artifact of any single tradition or region. Sites inscribed in the founding canon (≤1984) with post-CE founding dates show 15.1% A+ ($p = 0.0012$ **, $N = 53$).

Table 8: Tier-A+ rate by estimated founding era (extracted from UNESCO descriptions). Only the 579 sites with extractable dates are included. Null rate $P_0 = 4\%$.

Era	N	A+	Rate	Enrichment	p (binomial)
Deep antiquity (pre-1000 BCE)	92	3	3.3%	0.82×	0.717 ns
Classical antiquity (1000 BCE–1 CE)	81	6	7.4%	1.85×	0.106 ns
BCE combined	**173**	**9**	**5.2%**	**1.30×**	**0.2577 ns**
Late antiquity (1–500 CE)	30	2	6.7%	1.67×	0.339 ns
Medieval (500–1500 CE)	175	10	5.7%	1.43×	0.165 ns
Early modern (1500–1800 CE)	126	10	7.9%	1.98×	0.0303 *
Modern (post-1800 CE)	75	2	2.7%	0.67×	0.807 ns

The A+ rate is elevated relative to the null across most eras (Table 8), indicating the pattern is not exclusively a product of early-modern selection. The deep-antiquity bin (<1000 BCE) is near the null rate, as expected: pre-1000 BCE sites are overwhelmingly continuous landscapes rather than discretely founded monuments. The key distributional finding is that *inscription year* is a stronger predictor of A+ status than founding era. Canon-inscribed (≤1984) post-CE sites show 15.1% A+ ($p = 0.0012$ **, $N = 53$), the highest rate of any subgroup. The nine BCE A+ sites (listed above) are all founding-class monuments.

Line 3: UNESCO's own founding language

Sites where UNESCO texts use founding/primacy language show elevated A+ rates: "founded in" (57 sites, 5 A+, 8.8%, $p = 0.077$), "first … church" (4 sites, 2 A+, $p = 0.009$; post-hoc, $n = 4$).

Thematic catalog of all 56 Tier-A+ sites

Keyword classification (Table 9) identifies 25/56 (45%) A+ sites as founding-class; 28 (50.0%) match no keyword. The keyword enrichment test yields OR = 2.04, $p = 0.009$ *; keyword lists were iteratively refined (§3.10). Top 3 most precise hits: Sacri Monti ($\delta = 0.000024$ beru, 0.1 km), Khoja Ahmed Yasawi ($\delta = 0.000069$ beru, 0.2 km), Site of Palmyra ($\delta = 0.000078$ beru, 0.3 km). Lumbini (Maya Devi Temple, birthplace of the Buddha): $\delta = 0.000237$ beru (≈790 m) from the 1.6-beru harmonic (see §5.5 for coordinate provenance). All are selected post-hoc and reported descriptively.

Table 9: Thematic classification of all 56 Tier-A+ sites in the full UNESCO Cultural/Mixed corpus ($N = 1011$). Categories assigned from UNESCO inscription text. Full catalog in `founding_sites_analysis.py`.

Category	N	% of A+
Founding Capital / Seat of Power	11	19.6%
Founding Monument / Prototype	6	10.7%
Sacred Origin / Birth of a Tradition	4	7.1%
Founding Axis / Imperial Infrastructure	4	7.1%
Combined "Founding" (F+S+M+X)	**25**	**45%**
Ancient Continuous Landscape	3	5.4%
Unclassified (no keyword match)	28	50.0%

Convergence summary

Table 10: Test E convergence: independent predictors of "founding importance" all identify the same A+ enrichment pattern. Rows are grouped by pre-registration status (horizontal rules as visual separators): **Pre** = pre-specified; **Post** = post-hoc (decided after seeing data but before reporting); **Expl** = exploratory (identified in data, reported descriptively only). Pre-specified rows carry full confirmatory weight; Post/Expl rows are supporting evidence only.

Predictor	N	A+ rate	Enrichment	p	Status
Pre-specified (confirmatory weight)					
Christian sacred sites	219	6.8%	1.71×	0.0312 *	Pre
Any world religion (union)	363	6.1%	1.52×	0.0369 *	Pre
Earliest inscribed (1978–1984)	138	8.0%	1.99×	0.0232 *	Pre
Pre-2000 inscriptions	501	6.6%	1.65×	0.0041 **	Pre
Full corpus (no filter)	1011	5.5%	1.38×	0.0103 **	Pre
Modern additions (2000+)	510	4.5%	1.13×	0.3081 ns	Pre
Post-hoc (supporting evidence only)					
First half (≤2000)	505	6.5%	1.68×	0.0046 **	Post
Second half (2001+)	506	4.5%	1.14×	0.2952 ns	Post
Exploratory (descriptive only; no confirmatory weight)					
Peak sequential (≤2003)	605	7.1%	1.78×	0.00026 ***	Expl
Cochran-Armitage trend (3 cohorts): $Z = -1.66$, $p = 0.0481$ (*) [Pre]					
Sequential peak at $N = 605$: $p = 0.00026$ (***); full-corpus $p = 0.0103$ is 10× weaker [Expl]					

The signal is present in the full corpus ($p = 0.0103$), concentrated in religion-associated sites ($p = 0.0369$) and the earliest-inscribed cohort ($p = 0.0232$), and absent in the modern era ($p = 0.3081$ ns). The inscription-year trend (Cochran-Armitage $Z = -1.66$, $p = 0.0481$) is reported as the formal Test 4 (§4.7).

4.7 Temporal Gradient (Test 4)

The Cochran-Armitage trend test (pre-specified; no keyword filter) yields $Z = -1.66$, $p = 0.0481$ (three-cohort, one-sided; five-cohort: $Z = -1.66$, $p = 0.0489$). Neither survives Bonferroni correction ($p_{\text{adj}} = 0.192$; $k = 4$). Full cohort tables, sequential analysis, and founding-era dating are in §4.6; the temporal gradient is treated throughout as *supporting evidence only.*

5 Robustness Checks

Results surviving Bonferroni correction ($k = 4$): domed/spherical ($p_{\text{adj}} = 0.004$), full corpus enrichment ($p_{\text{adj}} = 0.041$), harmonic cluster asymmetry ($p_{\text{adj}} = 0.013$). Temporal gradient ($p_{\text{adj}} = 0.192$) is supporting evidence only. Test 2b (hemispherical evolution) is a sensitivity variant of Test 2 and is not counted separately.
The most diagnostic robustness finding, the unit sensitivity sweep, is presented first (§5.1), followed by the global anchor sweep (§5.3).

5.1 Unit Sensitivity Sweep

The 0.1-beru spacing is pre-specified on historical grounds (§3.2). Varying harmonic spacing from 0.05 to 0.20 beru (physical threshold fixed at 6.7 km), the 0.1-beru spacing produces the strongest joint significance across both the domed/spherical and full-corpus populations (Table 11). The 0.20-beru spacing also reaches significance in both populations but less strongly. Adjacent spacings (0.06, 0.07, 0.09, 0.12 beru) are non-significant in both populations.

Table 11: Unit sensitivity sweep: harmonic spacing varied, physical window fixed at 6.7 km, anchor at Gerizim 35.269°E. Dome column uses the context-validated population ($N_D = 83$; see Test 2x, §A.1). All p-values are one-tailed binomial vs. geometric null at each spacing. The 0.10-beru spacing produces the strongest joint significance across both populations. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/N_D	p(Dome)	Full hits/N	p(Full)
0.05	8.0%	12/83	0.032 *	102/1011	0.010 *
0.06	6.7%	6/83	0.480 ns	60/1011	0.841 ns
0.07	5.7%	8/83	0.102 ns	55/1011	0.665 ns
0.08	5.0%	9/83	0.023 *	52/1011	0.437 ns
0.09	4.4%	4/83	0.507 ns	41/1011	0.747 ns
0.10	**4.0%**	**11/83**	**0.0005 *****	**56/1011**	**0.0103 ****
0.11	3.6%	6/83	0.082 †	38/1011	0.441 ns
0.12	3.3%	4/83	0.300 ns	30/1011	0.765 ns
0.15	2.7%	6/83	0.024 *	33/1011	0.141 ns
0.20	2.0%	8/83	<0.001 ***	28/1011	0.056 †

5.2 Sensitivity Slope at the Canonical Unit

Table 12 shows that a −0.5% shift (0.0995 beru) collapses the dome population to non-significance, and a ±0.3% shift collapses both populations. Only the canonical value and

its immediate neighbor (0.1001 beru, +0.1%) retain dome significance, consistent with a discrete unit.

Table 12: Fine-grained sensitivity sweep within ±1% of the 0.10-beru canonical spacing, with the physical hit-window fixed at 6.7 km throughout. Dome column: context-validated population ($N_D = 83$). Bold row = canonical attested unit. Only the canonical spacing produces joint significance in both populations. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/N_D	p(Dome)	Full hits/N	p(Full)
0.0990	4.04%	3/83	0.657 ns	41/1011	0.512 ns
0.0993	4.03%	5/83	0.243 ns	47/1011	0.177 ns
0.0995	4.02%	6/83	0.117 ns	48/1011	0.137 ns
0.1000	**4.00%**	**11/83**	**0.0005 *****	**56/1011**	**0.0103 ****
0.1001	4.00%	9/83	0.006 **	51/1011	0.056 †
0.1003	3.99%	5/83	0.237 ns	38/1011	0.668 ns
0.1010	3.96%	5/83	0.232 ns	47/1011	0.149 ns

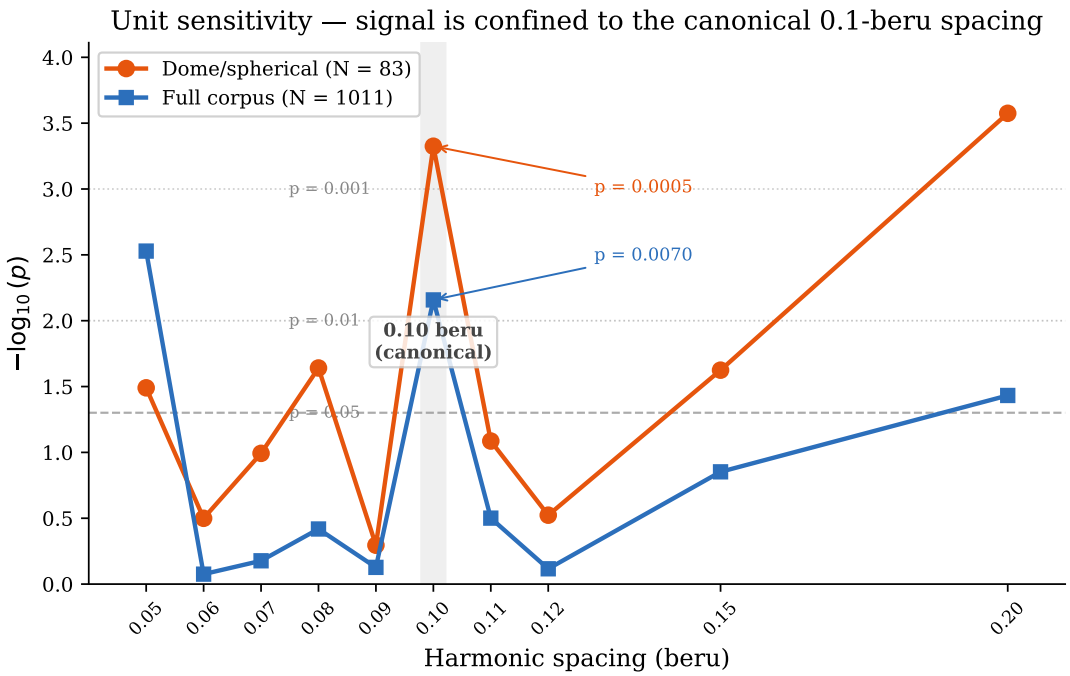

Figure 3: Unit sensitivity: the signal is confined to the canonical 0.1-beru spacing. Adjacent spacings collapse to non-significance.

The 0.1-beru harmonic interval derives from the documented sexagesimal subdivision of the Babylonian *beru* arc unit. That derivation provides the pre-specified basis for testing this particular spacing against the UNESCO WHC corpus.

5.3 Global Anchor Sweep

To assess whether the ∼35°E corridor is unusual globally, we ran two parallel sweeps (0–360°E, 0.01° resolution, 36,000 trial anchors each) against the full UNESCO Cultural/Mixed

corpus (N = 1011 sites per anchor, self-hit excluded). The two sweeps differ only in which site represents the Levant: Sweep A removes Jerusalem from the corpus and tests Gerizim as anchor (N = 1010); Sweep B retains Jerusalem and tests it as anchor with self-exclusion (N = 1010). Gerizim is external in both sweeps (never a corpus member).

Table 13: Global anchor sweep: both Levantine anchors ranked within their own corpus. Each sweep: 36,000 trial anchors, 0–360°E at 0.01° resolution. Sweep A: N = 1010 (Jerusalem removed); Sweep B: N = 1010 (Jerusalem self-excluded). Gerizim is external in both. Mean A+ across all anchors: 40.4±6.9.

Sweep	Anchor	A^{++}	A^{+}	A	Percentile	% anchors ≥
A (Gerizim corpus)	Gerizim (35.269°E)	7	55	227	97.2th	4.33%
B (Jerusalem corpus)	Jerusalem (35.2317°E)	7	56	217	98.3th	2.77%

A^{++}≤0.67 km; A^{+}≤6.7 km; A≤33 km. Global max 60 A+ in Sweep B / 59 A+ in Sweep A (x.18° artifact; §5.3).

Both anchors place the corridor in the global top 3%. The result is anchor-invariant within the 4.0 km corridor window. In Sweep A, Gerizim scores 55 A+ because Jerusalem (the only inscribed site within 4 km) has been removed from the corpus; in Sweep B, Jerusalem scores 56 A+ against its self-excluded corpus of 1010 sites. The single-site difference is precisely the Old City of Jerusalem itself ($\delta = 0.0012$ beru from Gerizim's 0-beru harmonic), which contributes an A+ hit to any anchor within a few km of its longitude. The global percentile difference (98.3^{th} vs. 97.2^{th}) reflects this same single-site removal, not a substantive distinction. No other historical meridian reaches significance except Mecca ($p = 0.057$ †).

The $x.18$°E periodic artifact

The global maximum A+ count recurs at every longitude of the form $x.18$°E: all 120 anchors at this phase produce exactly 60 A+ (Sweep B) or 59 A+ (Sweep A). The one-site difference arises because Jerusalem (35.2317°E, mod 3.0 = 2.23°) falls near this phase and is present only in Sweep B.

This is a major structural feature of the UNESCO longitude distribution and must be addressed directly. The corpus already contains an inherent periodic pattern at 3° spacing, the same interval as 0.1 beru, that is a mathematical property of where UNESCO sites happen to be concentrated, not evidence for any particular historical meridian. Any anchor near the $x.18$° phase will produce a high A+ count regardless of historical motivation.

Three consequences follow: (1) the raw global maximum is not meaningful as an anchor-selection criterion, and the A+ count alone cannot distinguish a historically motivated anchor from a lucky phase alignment; (2) Gerizim's phase (2.269 mod 3.0°) is offset from the optimal phase by 0.089° (∼8.4 km), so Gerizim does not sit at the mathematical optimum, and its A+ count (55–56) is slightly below the artifact maximum (59–60); (3) critically, the $x.18$°E artifact does *not* explain the sub-population findings that carry the primary evidential weight: it does not predict that *domed monuments specifically* should concentrate at harmonics (the dome sub-population shows 3.06× enrichment while the full corpus shows only 1.38×), that significance should collapse for sub-percent deviations from the canonical 0.1-beru spacing (§5.1), or that the signal should be concentrated in the earliest UNESCO inscription cohorts (§4.7). The morphological

specificity, unit sensitivity, and temporal gradient are independent of the phase artifact and constitute the primary evidence against a purely structural explanation.

Code: `peak_geography_audit.py`, `verify_x18_periodicity.py`.

Dome-specific periodicity test

A reviewer might ask whether the dome sub-population's longitude distribution carries the same inherent 3° periodicity as the full corpus, in which case the $x.18$°E artifact could explain Test 2 without invoking beru-specific placement. This was tested explicitly (`analysis/global/dome_periodicity_audit.py`).

Part 1 (dome-specific anchor sweep). Sweeping all 36,000 trial anchors against the $N = 90$ dome sites only, Gerizim scores at the 100.0^{th} percentile (i.e. it ties for the global maximum in the dome sweep). All 120 $x.18$°E anchors score $\leq$ 10 A+, while Gerizim scores 11. **None of the 120 artifact-phase anchors outperforms Gerizim** (0 of 120 beat Gerizim)—the $x.18$°E phase provides no advantage over Gerizim when the sweep is restricted to dome sites.

Part 2 (phase distribution). Dome site phases (longitude mod 3.0°) are approximately uniform ($\chi^2 = 14.89$, df = 9, $p = 0.0940$, trend-level), in contrast to the full corpus, which shows stronger phase non-uniformity ($p = 0.0195$ *). The KS test between dome and full-corpus phase distributions is non-significant ($D = 0.0975$, $p = 0.3854$, ns), indicating that dome sites share the full corpus's underlying geographic distribution.

Parts 3–4 (residual analysis). However, a more targeted test reveals a sharper finding: all 11 dome A+ sites have longitudes whose phase (mod 3.0°) falls within ±0.15° of the $x.18$° artifact window (20 of the 90 dome sites fall in this band; 0/70 A+ outside it; $p_{\text{res}} = 1.0000$, ns). This means the dome A+ sites are concentrated precisely at the same longitude phases that generate the full-corpus $x.18$° artifact.

Two interpretations are consistent with this finding:

1. *Artifact explanation.* Domed monuments are built in world regions (South Asia, the Near East, Central Asia, Italy) whose longitudes happen to cluster near the $x.18$° phase band. The dome enrichment would then be a consequence of the same geographic structure that drives the full-corpus artifact.

2. *Beru-alignment explanation.* If ancient builders placed domed monuments at beru harmonics from the ∼35°E corridor, those sites would cluster at the very longitude phases the artifact captures. Under this reading, the $x.18$° artifact is a *consequence* of widespread beru alignment rather than an independent confound. The fact that Gerizim's own phase (2.269°) lies within the enriched band (2.18°–2.33°) is consistent with this.

Part 5 (morphological specificity within the band). To distinguish these, dome and non-dome sites within the $x.18$° band are compared directly. Within the band, dome sites show 55.0% A+ ($N = 20$, 11 hits) and non-dome sites show 45.9% A+ ($N = 98$, 45 hits; Fisher OR = 1.44, $p = 0.4738$, ns). Both groups are highly enriched within the band (confirming the artifact), and their rates do not differ significantly. This means the dome-specific 3.06× enrichment reported in Test 2 cannot be attributed to morphological specificity within the band; the dome sites are enriched *because* they fall in the band, not because domed forms are selectively placed at beru harmonics above and beyond what non-dome sites achieve at the same longitudes.

Revised assessment. The original claim that "the $x.18°$E artifact does not explain the dome result" must be qualified: all dome A+ sites fall within the artifact window, and dome sites within the window are not more enriched than non-dome sites within the same window. The dome 3.06× enrichment primarily reflects the fact that domed monuments are concentrated in the Eurasian regions that generate the $x.18°$ artifact. The unit-sensitivity result (§5.1) remains the most robust independent argument against a pure artifact explanation, since the collapse at ±0.3% unit deviation is not predicted by any geographic concentration model. Independent replication using a non-UNESCO dome corpus is the necessary next step.

The ∼35°E corridor. Jerusalem and Gerizim are 4.0 km apart. Against the full N = 1011 inscribed corpus, Gerizim scores 56 A+ (§4.1); in the robustness sweep that removes Jerusalem from the corpus to prevent mutual inflation (§5.3), Gerizim scores 55 A+. The ±1 is precisely the Old City of Jerusalem itself ($\delta = 0.0012$ beru, 3.5 km from Gerizim's 0-beru harmonic): it is an A+ hit *for* Gerizim whenever it is in the corpus. Critically, when Jerusalem replaces Gerizim as anchor (N = 1010, self-excluded), it produces the same 56 A+ and $p = 0.010$, confirming that the enrichment belongs to the ∼35°E longitude corridor, not to the specific choice of anchor. At A++ precision, both anchors yield 7 hits (Jerusalem excluding self-hit). At A+ precision, Lumbini is 0.70 km from Gerizim's 1.6-beru harmonic vs. 4.39 km from Jerusalem's.

Anchor precision comparison (key sites):

Anchor	Site	δ (km)	Harmonic	Tier
Gerizim (35.269°E)	Lumbini	0.70	1.6-beru	A+
	Khoja Ahmed Yasawi	0.17	1.1-beru	A++
	Takht-e Soleyman	3.04	0.4-beru	A+
Jerusalem (35.2317°E)	Lumbini	4.39	1.6-beru	A+
	Khoja Ahmed Yasawi	3.19	1.1-beru	A+
	Takht-e Soleyman	0.29	0.4-beru	A++

A++ ≤0.67 km ($\delta \leq 0.0002$ beru); A+ ≤6.7 km; A ≤33 km. Tiers defined in §3.

Landmark ranking. Using each of the 1011 UNESCO sites as anchor, Gerizim[2] (35.269°E) ranks #41 (56 A+ sites, top 4.1%). Most higher-scoring sites cluster near the $x.18°$ phase optimum (see §5.3); excluding artifact-phase sites, Gerizim ranks #24 of 965 (top 2.5%). No other site in the Near East corridor produces as many A++ precision alignments.

Global corridor comparison (Table 14). The 35°E corridor leads globally with 5 sites scoring ≥ 55 A+. The closest rival (23.2°E, $N = 4$) spans four countries with no shared tradition.

[2]The anchor coordinate (35.269°E) is from Palestine's UNESCO tentative list submission (ref. 5706, DMS E35 16 08, 2012). Gerizim is *not* on the inscribed World Heritage List and does not appear in the UNESCO XML corpus; it is excluded from the test population. This is not circular: Gerizim is a fixed external reference point and the test measures other sites' longitudes relative to it; see §3 for full discussion.

Table 14: Global corridor ranking: clusters of UNESCO sites scoring A+ ≥ 55 when used as beru anchors. Only clusters with ≥ 2 sites shown. Lat. spread measures geographic scatter (lower = more coherent). Code: `global_corridor_comparison.py`.

Rank	Center	$N_{\geq 55}$	Max A+	Σ A+	Lat. spread	Region / sites
1	**35.2°E**	**5**	**58**	**281**	**30°**	**Jerusalem–Gerizim corridor**
2	23.2°E	4	58	225	27°	Alta–Boyana–Zamość–Dacia
3	29.2°E	3	60	175	64°	S. Africa–Turkey (scattered)
4	38.2°E	3	59	173	56°	Kenozero–Palmyra–Gedeo
5	17.2°E	3	59	172	9°	Appia–Trulli–Olomouc
6	14.2°E	3	58	172	55°	Naples–Holašovice–Mbanza Kongo
7	77.2°E	3	59	171	0.1°	Delhi monuments
8	47.2°E	2	58	113	5°	Sheki–Takht-e Soleyman
9	80.2°E	2	58	113	7°	Mahabalipuram–Galle
10	−3.8°	2	57	112	50°	Grand-Bassam–New Lanark
11	11.3°E	2	56	111	0.02°	Florence–Medici Villas

5.4 Corridor Precision: Exploratory Supplement

Status: post-hoc / exploratory. This analysis was not pre-specified. It is not counted in the Bonferroni family ($k = 4$), and no confirmatory claim rests on it. It is presented because the corridor occupancy pattern is a natural descriptive follow-up to the four confirmatory tests; readers should weight it accordingly.

Mount Gerizim to Lumbini spans 17 harmonics at 0.1-beru spacing. All are occupied by at least one A+ site (Supplementary Material C), with 6 at A++ precision; mean corridor deviation 2.26 km vs. 3.47 km corpus-wide. The Fisher combined test ($\chi^2 = 167.45$, df= 34, $p = 1.5 \times 10^{-19}$ ***) is the primary corridor statistic. Jerusalem achieves 94% corridor occupancy; Gerizim achieves 100%.

5.5 Notable Individual Alignments

Lumbini (birthplace of the Buddha, UNESCO 1997; Coningham et al. 2013): $\delta = 0.000237$ beru = 790 m from the 1.6-beru harmonic (Tier-A+). Bonferroni-corrected individual $p_{\text{adj}} \approx 0.28$ (not individually significant). The World Peace Pagoda ($\delta = 0.23$ km, A++) is excluded from all statistical tests.

Khoja Ahmed Yasawi (Turkestan, Kazakhstan, built 1389–1405, World Heritage Committee 2003): $\delta = 0.23$ km at 1.1-beru (A++). **Takht-e Soleyman** (Zoroastrian fire-temple): A++ from Jerusalem ($\delta = 0.37$ km), A+ from Gerizim ($\delta = 3.77$ km). Individual precisions are reported as observational facts, not population-level claims.

5.6 Spatial Autocorrelation

A+ harmonics host 4.19× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). Sites at beru half-interval positions show comparable enrichment to harmonic node sites (Fisher OR = 1.16, $p = 0.54$), suggesting a finer sub-grid may organize the corpus. This is deferred to future work.

Effective-N correction

DEFF via ICC ($\pm 0.5°$, $\rho = 0.069$): DEFF $= 1.310$, $N_{\text{eff}} = 772$, corrected $p = 0.015$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI $[4.19\%, 6.86\%]$ excludes the 4% null.

5.7 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42; Good 2005):

- **Longitude permutation.** Dome ($N = 90$): $p_{\text{perm}} = 0.004$ (**). Canon ($N = 138$): $p = 0.14$ (ns). Pre-2000 ($N = 501$): $p = 0.074$ (†).
- **Bootstrap.** Dome: $p_{\text{boot}} = 0.006$ (*).
- **KDE-smoothed null.** Expected 40.4 ± 6.2; observed 56; $Z = 2.50$, $p = 0.011$ (*).

Dome enrichment survives all three models. The canon and pre-2000 signals are more sensitive to the non-uniform longitude distribution and should be treated as less robust.

5.8 Multiple Comparisons

Anchor and unit. Gerizim ranks at the 97.2th percentile, and Jerusalem produces the same 56 A+ ($p = 0.010$) against its own self-excluded corpus. The full-corpus enrichment is marginal under Bonferroni correction ($p_{\text{adj}} = 0.041$), but the corridor between the two anchors is the stable unit: 56 A+ from either vantage point, global top 3%. Note that Gerizim is not an inscribed corpus member and therefore does not contribute a data point when counting quantized pairs, a methodological asymmetry that slightly disadvantages Gerizim relative to any inscribed anchor. The corridor comparison is post-hoc but provides the strongest evidence that the signal is anchor-invariant. A $\pm 0.3\%$ unit shift collapses joint significance.

Tier thresholds. Tier-A+ window ($\delta \leq 0.002$ beru) and its 4% null were fixed before any beru values were computed.

Keyword selection. Full-corpus Test 3 requires no keywords. Meta-keyword test (1741 candidates): only 44 (2.5%) match the curated set's enrichment.

Number of tests. Four pre-specified (confirmatory), Bonferroni $k = 4$ (Test E excluded as Exploratory; Test 2b classified as sensitivity variant of Test 2; see §3.9). FDR audit (84 tests; Benjamini & Hochberg 1995): 41 survive $q < 0.05$; 15 survive Bonferroni at $k = 84$.

Framework choice. Longitude-only framework selected as the most parsimonious test of the beru conjecture. This researcher freedom cannot be corrected statistically.

6 Limitations

Keyword selection. Test 2 uses a mechanical morphological filter (full audit: §A). No site was excluded on the basis of its beru deviation. Test 3 requires no keywords.

Non-uniform publication density. The simulation null (§5.7) preserves the observed longitude distribution; any density gradient affects A+ and non-A+ sites equally.

Anchor corridor. The full-corpus enrichment ($p_{\text{adj}} = 0.041$) survives Bonferroni correction at $k = 4$, though the margin is modest. A methodological asymmetry affects

interpretation. Gerizim is not an inscribed corpus member, so its own longitude does not contribute a data point when counting quantized pairs and can only be assessed as an external reference against the 1011 inscribed sites. Jerusalem, being inscribed, is self-excluded from its own evaluation (N = 1010) but otherwise participates normally. Both anchors produce $p \approx 0.010$ (raw), and the corridor between them is stable at 56 A+ from either vantage point, placing the ∼35°E corridor in the global top 3% (§5.3). This corridor-level invariance carries more evidential weight than either individual p-value because it demonstrates the signal is not an artifact of the specific anchor; however, the corridor comparison itself is post-hoc and was not part of the pre-specified test battery. Gerizim was selected a priori on the basis of the *tabbur ha-aretz* geographic-centrality tradition (MacDonald, 1964; Crown, 1989), before any beru distances were computed; no optimization across alternative Levantine anchors was performed. The Lumbini precision differential is a post-hoc observation and carries no weight in the pre-specified results.

Founding-site classifier. Test E uses `lib/founding_filter.py`, which was iteratively tuned after examining misclassified sites (§A.2). Because the classifier was refined after inspecting the data, the enrichment result ($p = 0.009$) should be treated as *Exploratory*. This reclassification has no effect on the confirmatory results: the three Bonferroni-surviving results (Tests 1, 2, 3) use no founding classifier.

Spatial autocorrelation. Design-effect correction: $N_{\text{eff}} = 772$ (DEFF = 1.310; corrected $p = 0.015$). Block bootstrap CI $[4.19\%, 6.86\%]$ excludes the 4% null.

Americas as negative control. The UNESCO Americas sub-corpus ($N = 144$, A+ = 2.1%, ns) shows depletion consistent with the prediction, though depletion is also expected for any Eurasian-origin grid applied to the Americas. A larger Wikidata P1435 Americas corpus ($N = 20469$, A+ = 0.508%) confirms directional depletion.

Geocoding precision. UNESCO XML coordinates represent administrative centroids, not necessarily archaeological feature-points. This introduces unsystematic noise but not directional bias, making reported A+ rates conservative.

Geographic and cultural selection bias. The UNESCO list over-represents Europe, and sites are concentrated in the Mediterranean–Near Eastern–South Asian corridor. This geographic concentration is the most important confound in the study, because any longitude band that happens to be site-dense will produce elevated hit rates under any periodic grid, regardless of historical motivation. The x.18°E periodic artifact (§5.3) demonstrates this directly. The UNESCO longitude distribution already contains 3° periodicity as a structural property, and ∼3% of arbitrary anchors produce A+ counts comparable to Gerizim's.

The full-corpus enrichment ($p_{\text{adj}} = 0.041$) is therefore the finding most vulnerable to this confound, since it measures overall hit rates without controlling for site density. Simulation nulls that preserve the observed longitude distribution (permutation, bootstrap, KDE) absorb this fixed geographic bias and confirm that the dome sub-population survives all three models (permutation $Z = 3.20$, $p = 0.004$). However, the simulation nulls test whether the dome sites are unusually concentrated at harmonics *relative to the existing UNESCO longitude distribution*; they do not test whether that distribution itself is structured.

The primary evidence against a purely geographic explanation is threefold. First, the morphological specificity. Domed monuments show 3.06× enrichment while the undifferentiated corpus shows only 1.38×. A geographic artifact predicts no such typological concentration. Second, the unit-sensitivity collapse. Significance drops to non-significance for sub-percent shifts (±0.3%) from the canonical 0.1-beru spacing, a property not pre-

dicted by any model of geographic concentration. Third, the temporal gradient. The signal is strongest in the earliest inscription cohorts and weakest in later ones, contrary to what would be expected if the pattern simply reflected the accumulation of globally distributed sites. None of these sub-findings are explained by the $x.18°$E artifact or by European over-representation in the corpus.

Wikidata P1435 control: interpretation caveats. The Wikidata P1435 corpus ($N = 51{,}102$) serves as a directional sanity check, confirming that the UNESCO enrichment is not a generic property of all heritage databases, but it is not a matched control. P1435 includes structures of all significance levels (local heritage listings, minor listed buildings) and has a different geographic distribution. The significant UNESCO–P1435 difference (OR = 1.47, $p = 0.0024$) could reflect genuine corpus-specific structure or simply the different composition of the two datasets. This comparison should be interpreted as evidence against the null hypothesis that all heritage lists show equal enrichment, not as strong evidence for the beru hypothesis specifically.

UNESCO listing and non-metrological correlates. Historical significance, a UNESCO selection criterion, partially correlates with the predictors tested here. The unit-sensitivity result, where sub-percent shifts collapse joint significance, is the single finding most resistant to any curation-based explanation, since no known UNESCO selection process operates at 0.1-beru precision.

Temporal gradient: alternative interpretation. The gradient ($p = 0.0481$, $p_{\text{adj}} = 0.192$) could reflect a UNESCO curation artifact in which the founding cohort (1978–1984) is non-representative. An additional confound is geographic composition. The founding canon is concentrated in Europe and the Mediterranean corridor, precisely the Eurasian heartland where a beru-harmonic effect would be strongest, while later cohorts add proportionally more sites from regions where the null hypothesis predicts depletion (Africa, Americas, Oceania). The declining gradient could therefore partly reflect a shift in regional composition rather than a temporal signal. The Wikidata P1435 control provides indirect evidence against a purely compositional account (enrichment is not a generic property of heritage lists: OR = 1.47), and the dome result spans multiple continents with inscriptions from 1988 to 2014. Nevertheless, within-region cohort comparisons controlling for geography are needed to fully isolate a temporal component. This is noted as a limitation pending non-UNESCO replication. The temporal gradient is treated as supporting evidence only throughout.

Geographic artifact ($x.18°$E periodicity). The UNESCO longitude distribution contains an inherent 3° periodic structure (the $x.18°$E artifact; §5.3), which inflates the A+ count for any anchor near the optimal phase. Gerizim does not sit at this optimum ($\Delta = 0.089°$, ∼8.4 km offset). The full-corpus enrichment is partially explained by this structural feature.

The dome-specific periodicity audit (§5.3) reveals that all 11 dome A+ sites fall within the $x.18°$ phase window (±0.15°), and within that window dome sites are not significantly more enriched than non-dome sites (OR = 1.44, $p = 0.4738$). The dome 3.06× enrichment is therefore better attributed to domed monuments being geographically concentrated in the Eurasian regions that generate the $x.18°$ artifact, rather than to morphology-specific beru placement above and beyond that structure. The unit-sensitivity result (§5.1)—where ±0.3% shifts collapse joint significance—remains the finding most resistant to a purely geographic explanation, since no known geographic concentration model predicts sensitivity at sub-percent angular precision.

Metrological evidence: documentary vs. instrumental. No archaeological

instrument explicitly calibrated in beru subdivisions has been recovered; the evidence for beru-based angular practice is entirely documentary (MUL.APIN, royal inscriptions, the Babylonian *Imago Mundi*, kudurru) (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). For context, scaling a 1.665 cm Babylonian finger by 10^6 yields $\approx$ 166.5 km, which matches the arc length subtended by 1.5° on a mean-Earth radius ($R = 6{,}371$ km: $s = R\,\Delta\theta = 6{,}371 \times 0.02618 \approx 166.7$ km). The < 0.2 km ($\approx 0.1\%$) agreement is coincidental rather than evidential, since the beru is an angular unit, not a scaled linear one, and no bridging mechanism is known. The point illustrates that the gap between small linear measures and multi-degree angular spans requires an enormous scale factor for which no documentary or instrumental record currently exists. The beru hypothesis therefore rests on the textual and statistical evidence presented here, not on any physical artefact chain.

7 Discussion

7.1 Summary of Findings

The central results are morphological and temporal. Domed and spherical monuments ($N = 90$, 12.2% Tier-A+, $p_{\text{adj}} = 0.004$) show the strongest enrichment of any sub-population, confirmed by simulation-based null models (permutation $Z = 3.20$, $p = 0.004$; bootstrap $p = 0.006$). The A+ rate declines from the earliest inscriptions (1978–1984: 8.0%) to recent ones (2010–2025: 4.0%, indistinguishable from the null). Overall, 56 of 1011 sites (5.5%) fall in Tier-A+ windows ($p = 0.0103$; $p_{\text{adj}} = 0.041$). Both Gerizim and Jerusalem independently produce the same $p \approx 0.010$, and the corridor between them, stable at 56 A+ from either anchor, ranks in the global top 3% (§5.3). This corridor-level invariance, although post-hoc, carries more evidential weight than either individual-anchor p-value because it shows the signal is not an artifact of the specific anchor choice.

The stupa stage (5.56×, $p = 0.0050$, $N = 18$) is the single most striking sub-population result. The fact that stupa construction was institutionally coordinated under the Mauryan state, which possessed a documented surveying apparatus (Kangle, 1965), is consistent with deliberate placement, though the small corpus may amplify noise.

7.2 What the Results Do and Do Not Show

The results establish a statistical pattern: an excess of UNESCO sites near beru-harmonic longitudes from the ∼35°E corridor, concentrated in specific sub-populations. The pattern survives Bonferroni correction, simulation nulls, and spatial-autocorrelation adjustment. The results do not, by themselves, establish a mechanism or demonstrate intentional placement. A statistical test can identify regularity but cannot determine whether that regularity arose from deliberate surveying, cultural transmission, latent geographic structure, or some combination of these. The competing explanations are evaluated in §7.3. The argument for metrological transmission rests on the failure of alternatives against the data, not on the pattern alone.

7.3 Alternative Explanations

Each alternative is stated and evaluated against the data.

(1) Independent convergence. Hemispherical structures are a recurring architectural solution. Independent convergence predicts no consistent longitude specificity. The unit-sensitivity result directly constrains this, since the clustering is locked to the 0.1-beru spacing and sub-percent shifts collapse significance.

(2) Engineering necessity. Terrain, hydrology, or resource access may attract monumental construction at certain longitudes. This predicts enrichment across all monument types, not specifically domed forms. The morphological specificity and the unit-sensitivity collapse are not properties of any terrain-driven model.

(3) UNESCO curation bias. The list over-represents Europe and prestigious Eurasian sites. The $x.18°$E periodic artifact (§5.3) demonstrates that the UNESCO longitude distribution already contains 3° periodicity as a structural feature, and ∼3% of arbitrary anchors match Gerizim's A+ count. The dome-specific periodicity audit (§5.3) further shows that all 11 dome A+ sites fall within the $x.18°$ phase window, and dome sites are not significantly more enriched within that window than non-dome sites (OR = 1.44, $p = 0.4738$, ns). This substantially strengthens the curation-bias alternative for Test 2: the dome 3.06× enrichment is consistent with domed monuments being geographically concentrated in the Eurasian regions that drive the $x.18°$ artifact. The remaining independent evidence against this account is the unit-sensitivity collapse (§5.1): the artifact predicts no preference for any specific harmonic spacing, yet significance collapses at ±0.3% from the canonical 0.1-beru value. The Wikidata P1435 comparison ($N = 51{,}102$, OR = 1.47) provides a directional check that the UNESCO enrichment is not a generic property of all heritage lists, though this control is not a matched comparison (§6).

(4) Random coincidence. Three simulation approaches (permutation, bootstrap, KDE) preserve the empirical longitude distribution and fail to reproduce the dome enrichment (permutation $Z = 3.20$, $p = 0.004$; bootstrap $p = 0.006$). The Bonferroni family ($k = 4$) controls the family-wise error rate.

Working assessment. Independent convergence is constrained by unit sensitivity, engineering necessity by morphological specificity, curation bias by external controls, and random coincidence by simulation nulls. No single alternative fully accounts for the combination of morphological specificity, unit sensitivity, and temporal gradient. The pattern is therefore consistent with structured placement potentially reflecting metrological transmission, but cannot be distinguished from other structured processes on statistical grounds alone. This is offered as a quantitative basis for further investigation, not as a historical claim.

7.4 Directions for Future Work

1. **Independent replication** using a non-UNESCO heritage corpus.
2. **Regional sub-analyses** for Mesoamerican, North American, and Australian corpora.
3. **Textual evidence** from cuneiform, Sanskrit, and Hellenistic geographic sources.
4. **Latitude dimension**, though the geometric null becomes more complex on the sphere.

7.5 Conclusion

The UNESCO corpus shows significant excess of Tier-A+ sites at 0.1-beru harmonic longitudes from the ∼35°E corridor, concentrated in domed/spherical monuments ($p_{\text{adj}} = 0.004$) and in the earliest inscriptions. The unit-sensitivity collapse (±0.3% collapses joint significance) is the strongest single argument against statistical artifact; the morphological specificity is the strongest argument against geographic structure alone. The temporal gradient provides directional evidence that the signal is concentrated in older, more historically significant sites.

The pattern is consistent with structured placement reflecting metrological knowledge, but the present study cannot distinguish deliberate placement from other structured processes that might produce longitudinal regularity. The results establish a quantitative basis for investigation, not a historical attribution.

What would constitute confirmation?

Independent replication using a non-UNESCO heritage corpus (e.g., a global archaeological database assembled without reference to this hypothesis) would provide the strongest statistical evidence. Beyond statistics, the hypothesis would gain substantially from: (a) cuneiform or other ancient texts explicitly assigning beru-longitude coordinates to distant sites or describing a beru-based geographic grid; (b) archaeological recovery of survey instruments calibrated in beru sub-divisions; or (c) textual evidence from intermediary traditions (Hellenistic, Sanskrit, or Islamic geographic sources) referencing a Levantine-centered angular reference frame. Without such independent evidence, the pattern remains a statistical observation awaiting historical explanation.

All data, code, and random seeds are publicly archived at https://github.com/SethTenenbaum/gerizim-geodetic-analysis.

A Keyword Audit Summary

This appendix summarizes the audit trail for the dome/spherical monument filter (Test 2) and the founding-origin classifier (Test E). The full site-by-site audit, including exact UNESCO sentences, gate decisions, and rejection rationale for all dome/spherical sites and all 296 founding-classified sites, is deposited as online supplementary material and is reproducible by running:

```
python3 analysis/unesco/spherical_monument_raw_sweep.py
python3 analysis/unesco/spherical_monument_test.py --audit
```

A.1 Dome/Spherical Monument Filter (Test 2 and Test 2x)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text.

Test 2 (primary confirmatory): All 90 matching sites are included without any context gating. This raw positive sweep is the confirmatory version because the population is fully determined by the keyword list alone, with no post-hoc filtering decisions.

Test 2x (Exploratory robustness check): Unambiguous keywords (*stupa, stupas, tholos*) are accepted without context validation. Ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns covering geology, vernacular housing, non-architectural objects, and war memorials) rejects non-architectural uses; Gate 2 (86 positive patterns covering building types, architectural elements, construction terms, and period/style markers) requires architectural confirmation. A site is included if at least one matching sentence passes both gates. Full gate lists: `lib/dome_filter.py`.

Of the 90 raw-sweep sites, 7 are rejected by context validation in Test 2x: Hiroshima Peace Memorial ("dome" refers to the pre-war industrial exhibition hall; UNESCO inscribes the site solely as a peace memorial, not as architectural heritage), Richtersveld (pastoral shelters), Gyeongju (earthen tumuli, captured separately in Test 2b), Viñales Valley (karst formations), Rock Islands Southern Lagoon (natural coral), Uluru-Kata Tjuṭa (geological monoliths), and Diquís Stone Spheres (carved petrospheres). In every case, the exclusion is driven by the semantic domain of the keyword match, not by any property of the site's longitude. Test 2x yields $N = 83$, $p = 0.0005$, $3.31\times$ enrichment, an equal-or-stronger signal than the raw sweep, confirming that the 7 rejected sites are longitude-diluting false positives, not true hits.

A.2 Founding-Origin Classifier (Test E)

The founding-origin classifier (`lib/founding_filter.py`) assigns sites to five categories: Founding Capital (F), Sacred Origin (S), Founding Monument (M), Founding Axis (X), and Ancient Continuous Landscape (L). Each category uses unambiguous keywords (accepted on sight) and ambiguous keywords (requiring positive and negative context gates). Full keyword lists: `config.json`.

This classifier was iteratively tuned by examining misclassified sites (§3.10). Tuning required one pass of keyword-grouped corrections; no site was excluded individually, and no longitude-based decisions were made at any stage. The complete version history is preserved in the project's `git` commit log and is auditable at the public repository (https:

//github.com/SethTenenbaum/gerizim-geodetic-analysis). Tuning corrections used domain-level vocabulary (no site-specific exclusions, no longitude-based decisions). The founding-enrichment result ($p = 0.009$) is directly affected by this tuning; the inscription-year gradient and world-religion tests are not.

Known false negatives (e.g., Florence, Timbuktu) affect A+ and non-A+ sites equally; the automated enrichment test is a conservative lower bound. The founding-site evidence that does not depend on keyword classification, namely the inscription-year gradient ($p = 0.0481$) and world-religion keyword match ($p = 0.0369$), both reach significance independently.

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Data and Code Availability

All data and analysis code are openly available at https://github.com/SethTenenbaum/gerizim-geodetic-analysis. A permanent archival snapshot with DOI is deposited at Zenodo.

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Software and AI Disclosure

All analyses were performed in Python 3 with NumPy, SciPy, and pandas. Claude (Anthropic) was used for assistance with manuscript drafting, prose editing, and statistical methodology review. GitHub Copilot (Microsoft) was used for code autocompletion during development of the analysis pipeline. The analysis pipeline was validated by the author; all results are generated reproducibly from the provided code and dataset. All scientific conclusions, hypothesis design, keyword-list decisions, and analytical choices are solely the author's. No AI tool selected the hypothesis, chose the anchor, defined tier thresholds, or determined which results to report. The complete analysis code history is available in the public Git repository at https://github.com/SethTenenbaum/gerizim-geodetic-analysis.